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OVERCOMING THE UNCERTAINTY CHALLENGES IN FLOOD RAPID MAPPING WITH MULTI-SOURCE OPTICAL DATA

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ABSTRACT

As global climate change worsens, floods are becoming more severe and frequent, urgently demanding effective flood risk mitigation strategies. Timely and precise flood inundation mapping is crucial for emergency response and relief. The 2024 IEEE GRSS Data Fusion Contest Track 2 aims to pioneer innovative algorithms for accurate flood extraction using multi-source optical remote sensing (RS) data. However, data diversity introduces aleatoric uncertainty, especially with synthetic, non-real data. Meanwhile, the vast coverage of RS imagery and the small proportion of flood areas cause a significant class imbalance, leading to epistemic uncertainty. In this paper, we propose an Uncertainty-aware Detail-Preserving Network (UADPNet) for rapid flood mapping of multi-source optical data. Firstly, we design an Aleatoric Uncertainty Estimator to model aleatoric uncertainty in multi-source data. Secondly, we introduce a Multi-Scale Convolution Block to extract multi-scale information without downsampling. Thirdly, we utilize a multi-level supervised strategy to quantify epistemic uncertainty and highlight uncertain pixels via the Uncertainty-Aware Fusion Module. With UADPNet, we adopt a multi-model fusion and post-processing strategy to enhance the final flood extraction accuracy. Our outstanding experimental results in the official test set showcase the superiority of our method, which secured the first-place ranking in the 2024 IEEE GRSS Data Fusion Contest Track 2, boasting an impressive F1 score of 89.843%.

***Index Terms*—** multi-source optical data, flood mapping, semantic segmentation, uncertainty-based theory

1. INTRODUCTION

Flood disasters, among the most frequent natural calamities, pose a dire threat to national economies and the safety of lives and properties. Consequently, flood extraction has garnered significant attention in natural disaster research. Traditional methods of flood information extraction, including field investigations [1] and manual mapping [2], are notoriously time-consuming and labor-intensive, often falling short in meeting the demands of extensive coverage and real-time monitoring. Therefore, there is a pressing need to explore more efficient and accurate techniques for flood detection and monitoring.

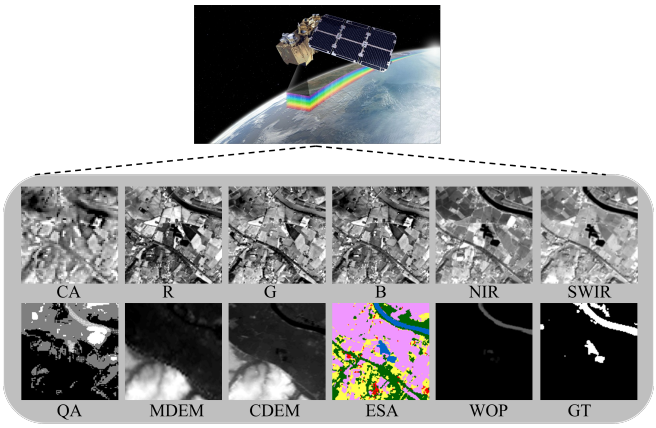


Fig. 1: Illustration of multi-source remote sensing (RS) images used in the DFC Track2.

The 2024 IEEE GRSS Data Fusion Contest Track 2 focuses on developing algorithms that can generate flood maps by integrating spatial data from diverse sources. Fig. 1 illustrates the various types of data utilized in this contest, including optical remote sensing (RS) images (Blue, Green, Red, NIR, SWIR1, SWIR2), quality assessment (QA) guidance, coastal aerosol (CA) information, digital elevation models (including Merit DEM and Copernicus DEM), cover map (ESA World Cover Map [3]), and water occurrence probability (WOP). The data for the competition is sourced from flood events annotated by the Copernicus Emergency Management Service, the OPERA Dynamic Surface Water Extent CalVal database, and CERFACS simulations of flooded areas corresponding to Copernicus Sentinel acquisitions.

However, the diverse array of data sources inevitably introduces aleatoric uncertainty, arising primarily from the inclusion of synthetic and non-authentic data. Additionally, the vast coverage of RS imagery, coupled with the relatively small proportion of flood-affected areas, creates a significant imbalance between foreground and background classes, further exacerbating epistemic uncertainty [4]. Moreover, in

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complex flood scenarios, the limited resolution of 30 meters poses challenges in discriminating between similar features, significantly increasing epistemic uncertainty in the models [4]. These factors pose significant challenges in accurate flood detection and analysis.

To address these uncertainties, we propose an Uncertainty-aware Detail-Preserving Network (UADPNet) to achieve rapid flood mapping for multi-source optical data. Specifically, we introduced an Aleatoric Uncertainty Estimator which employed the conditional variational autoencoder technology to accurately estimate and constrain the consistency between the prior and posterior distributions of noise, thereby obtaining aleatoric uncertainty at the data level. Subsequently, due to the low resolution of the data ($30m$) and limited input image size (128×128), we have abandoned the downsampling operation commonly used in previous semantic segmentation methods [5, 6, 7, 8, 9, 10, 11, 12] for feature extraction. Instead, we designed a simple and efficient Multi-Scale Convolution Block to construct a size-invariant context-aware encoder to extract multi-level encoded features. In the decoding process, we have developed a comprehensive flood mapping solution to solve the epistemic uncertainty of the model. We used a multi-level supervised strategy to quantify epistemic uncertainty and highlight uncertain pixels via the Uncertainty-Aware Fusion Module, generating low-uncertainty flood extraction results.

Utilizing the proposed UADPNet, we adopted a five-fold cross-validation strategy, randomly dividing the data into five distinct sets for model training and validation. Subsequently, we integrated the optimal outcomes from these five models through a multi-modal fusion approach. Furthermore, to derive the ultimate flood extraction results, we conducted a union operation on the water body labels within the land cover data. Notably, our experiments on the official test set yielded an impressive $F1$ score of 89.843%, underscoring the effectiveness of our proposed methodology.

2. METHODOLOGY

2.1. Stage A: Aleatoric Uncertainty Estimator

In the realm of flood extraction, our primary objective is to minimize the overall loss function, which serves as a metric for evaluating the model's performance. This loss function is formally expressed as:

$$\begin{aligned} &\min_{\theta} \mathbb{E}_{X,Y}[\mathcal{L}(f(X;\theta),Y)] \\ &\approx \frac{1}{N}\sum_{i=1}^{N}\mathcal{L}(f(x_i;\theta),y_i), \end{aligned} \tag{1}$$

where θ denotes the learned parameters by the model $f(\cdot)$, and (x_i, y_i) are individual samples drawn from the joint data distribution $p(X, Y)$, where X represents the concatenation of the NIR, Green, and Blue bands from the dataset, and Y refers to the corresponding ground truth. The function $\mathcal{L}(\cdot)$ quantifies the loss between the model's predictions and the ground truth.

As mentioned earlier, due to the presence of simulated data in the training dataset, coupled with a certain degree of human error in the real observation data, there is an inevitable uncertainty (aleatoric uncertainty) at the data level. To address this challenge, we utilize the Conditional Variational Autoencoder (CVAE [13]) to estimate this aleatoric uncertainty using maximum likelihood estimation. In CVAE, the prior distribution of latent variables is conditioned on the input data X and follows a Gaussian distribution. A typical conditional generative model comprises three key components: conditional variables X, latent variables z, and output variables Y. The latent variable is defined by the conditional distribution $P_\theta(z|X)$, and given the joint input conditional variable X, the distribution of the output variable Y, is obtained as $P_\omega(Y|X, z)$. Additionally, the posterior distribution of z is represented as $Q_\phi(z|X, Y)$. The loss function of CVAE is carefully designed to capture both the reconstruction accuracy and the closeness of the learned latent space to the true prior distribution. It is defined as:

$$\begin{aligned} L_{CVAE} = &\mathbb{E}_{z\sim Q_\phi(z|X,Y)}[-\log P_\omega(Y|X,z)] \\ &+\mathbb{D}_{\mathrm{KL}}(Q_\phi(z|X,Y)||P_\theta(z|X)), \end{aligned} \tag{2}$$

Here, $P_\omega(Y|X, z)$ represents the likelihood of observing the output variable Y given the latent variable z and the conditional variable X. The second term, $\mathbb{D}_{\mathrm{KL}}(Q_\phi(z|X, Y)||P_\theta(z|X))$, quantifies the closeness between the learned posterior distribution $Q_\phi(z|X, Y)$ and the prior distribution $P_\theta(z|X)$ using the Kullback-Leibler divergence. By minimizing this loss function, we aim to achieve both accurate reconstructions and a well-regularized latent space, enabling robust flood extraction even in noisy environments.

Adhering to the standard CVAE paradigm, we introduce an Aleatoric Uncertainty Estimator (AUE), whose components are meticulously detailed as follows. Precisely, in this study, we define $P_\theta(z|(X))$ as the prior network, which functions as a mapper, transforming the input image into a compressed latent space. Here, θ represents the parameter set specific to the prior network. Assuming identical network structures and the availability of ground truth Y, we designate $Q_\phi(z|(X), Y)$ as the posterior network, where ϕ comprises the parameters exclusive to the posterior network.

As illustrated in Fig. 2, within the hidden layer networks (both prior and posterior), we concatenate the multiple inputs and employ three 4×4 convolution blocks, each with a stride of 2. This process transforms the concatenated inputs of X (or X and Y in the case of the posterior network) into the latent variable z, which follows a normal distribution $N(\mu, \mathrm{diag}(\sigma^2))$. Here, μ and σ belong to $\mathbb{R}^K$ and represent the mean and standard deviation of the latent variable, respectively. This architecture not only preserves the essence

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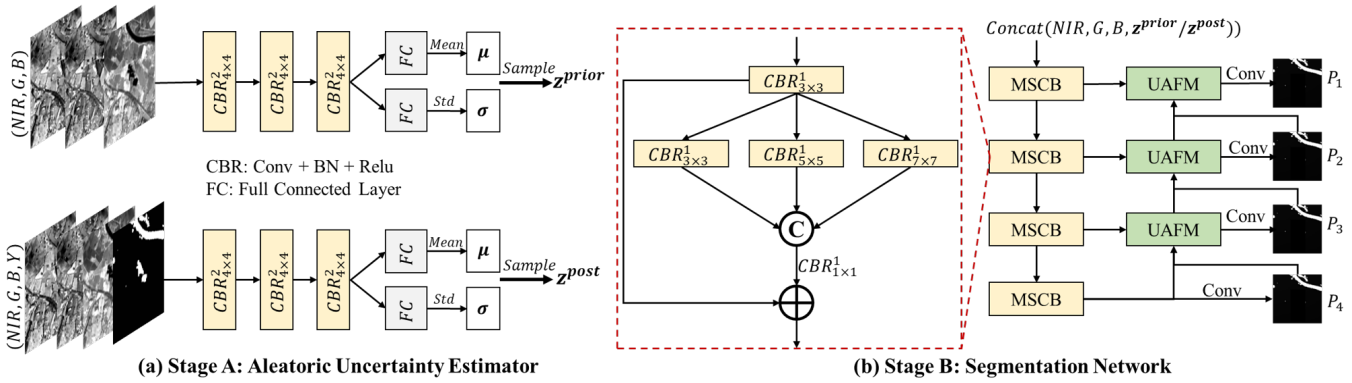


Fig. 2: The pipeline of our Uncertainty-aware Detail-Preserving Network (UADPNet), which is composed of two stages, *i.e.*, Aleatoric Uncertainty Estimator (AUE) and Segmentation Network.

of CVAE but also incorporates uncertainty awareness, rendering AUE a robust and versatile decoder.

2.2. Stage B: Segmentation Network

As depicted in Fig. 2, our segmentation network follows a U-shaped design akin to UNet. Specifically, we merge the estimated aleatoric uncertainty (EU) with X as the input to the entire network. Subsequently, we employ four Multi-Scale Convolution Blocks (MSCBs) to systematically extract four layers of encoded features.

The MSCB structure is straightforward: it first undergoes a 3×3 operation, followed by three parallel convolution branches (with convolution kernels of sizes 3×3, 5×5, and 7×7) aimed at capturing features with varied receptive fields. These branches are then subjected to dimensional reduction and restoration via a 1×1 convolution, supplemented by a residual connection to preserve the original feature information. This four-layer MSCB processing yields four layers of encoded features ($E_i, i = 1, 2, 3, 4$).

During the decoding phase, we take into account the challenges stemming from class imbalance in the dataset and the minor inter-class disparities in images caused by low resolution, resulting in model-level uncertainties (epistemic uncertainty). To tackle these issues, we employ a progressive decoding strategy utilizing our Uncertainty-Aware Fusion Module (UAFM) introduced in UANet [14]. We leverage the uncertainty evident in the preliminary flood extraction results to guide the model's attention towards challenging samples with higher uncertainty during training. This progressive methodology aims to systematically reduce uncertainties and accomplish precise flood extraction.

Specifically, we first use a 1×1 convolution directly to process E_4 and output P_4. Subsequently, we adopt the uncertainty ranking algorithm from UANet [14] to measure the uncertainty level of each pixel in the feature space and allocate different weights based on the measured uncertainty level. In detail, we directly use the $Sigmoid$ function to get the corresponding probabilities of all pixels in the extraction map P_4 from spatial perspective, then we subtract all values of the probability map with 0.5 to measure the uncertainty belonging to foreground flood (U_f) and meanwhile subtract 0.5 with all values of the probability map to measure the uncertainty belonging to background (U_b),

$$\begin{aligned} U_f &= Sigmoid(P_4) - 0.5, \\ U_b &= 0.5 - Sigmoid(P_4). \end{aligned} \tag{3}$$

Subsequently, we rank the uncertainty of the foreground and background into five levels using the Uncertainty Rank Algorithm (URA), which is described as:

$$\mho(i,j) = \begin{cases} \lfloor \frac{0.5 - U_{i,j}}{0.1} \rfloor, U_{i,j} >= 0, \\ 0, U_{i,j} < 0, \end{cases} \tag{4}$$

where $U_{i,j}$ means the pixel in i_{th} row and j_{th} column of U_f or U_b.

Afterward, we apply URA to P_4 so that we can get the corresponding foreground uncertainty rank map (R_f^4) and background uncertainty rank map (R_b^4). Then we directly use E_4 and E_3 to multiply with them to highlight the uncertain pixels from both the foreground and background perspectives. Subsequently, we concatenate these enhanced features and recover its original channel to get G_3 by a 1×1 convolution operation, which can be described as:

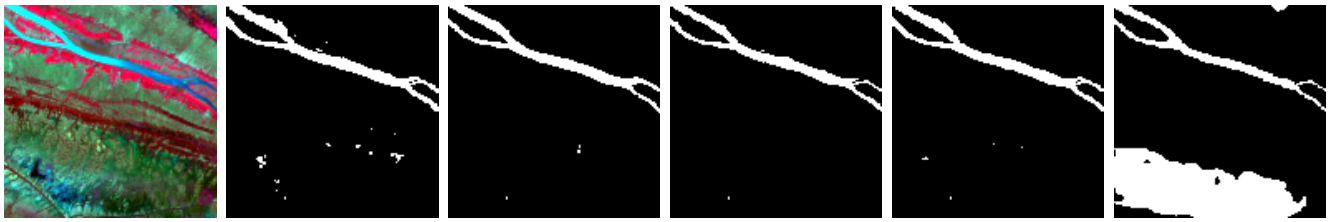
$$G_3 = Conv_{1\times1}(C(R_f^4 * E_4, R_b^4 * E_4, R_f^4 * E_3, R_b^4 * E_4)), \tag{5}$$

where C is short for Concatenation and G_3 is processed by 1×1 convolution to output P_3 with less uncertainty than P_4. With such a UAFM, we can utilize P_4 to fuse E_4 and E_3 and achieve output G_3 and P_3, utilize P_3 to fuse G_3 and E_2 and achieve output G_2 and P_2, and utilize P_2 to fuse G_2 and E_1 and output P_1, where P_1 can be viewed as the final refined extraction map with the lowest uncertainty.

2.3. Loss Function

On the whole, we use the simple binary cross-entropy (BCE) loss function to supervise all the extraction outputs and use

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(a) RGB Image (b) Deeplab v3+ [15] (c) HRNet [16] (d) UNet [17] (e) SegFormer [18] (f) Ours

Fig. 3: Visual Comparison between our proposed framework and the compared methods.

Table 1: Quantitative results on the official test dataset.

Method	Test (%) $F1 \uparrow$
UNet	82.093
HRNet	80.790
Deeplab v3+	80.697
SegFormer	83.426
UNet (MSCB)	84.507
UNet++ (MSCB)	84.577
Attention UNet (MSCB)	86.033
MSCB+UAFM	87.027
MSCB+UAFM+AUE	87.170
Multi-Fusion	<u>89.272</u>
Multi-Fusion + Post-Processing	**89.843**

L_{CVAE} to supervise the closeness of the learned latent space to the true prior distribution in AUE. The overall loss is:

$$Loss = \sum_{i=1}^{4} BCE(P_i, GT) + \lambda \cdot L_{CVAE}, \qquad (6)$$

where λ is 0.01.

3. EXPERIMENTS

3.1. Experimental Settings

For a comprehensive evaluation of our proposed model, all experiments were conducted using PyTorch 1.8.1 (CUDA 11.1) on an NVIDIA GeForce RTX 3090 GPU with 24GB of memory. During the training phase, we employed the AdamW [19] optimizer and applied the cosine annealing strategy to adjust the learning rate, starting from an initial learning rate of 10^{-4}. We adopted a five-fold cross-validation approach to determine the final model and utilized a multi-model fusion strategy to improve the extraction performance. Simultaneously, upon analyzing the official training data, we discovered the crucial role of provided land cover data in post-processing. To enhance the post-processing process, we merged the predicted results of our strategy on the test set with the designated water regions from the land cover data. It is worth noting that we applied twelve enhancements to each image in the training set by using six rotation angles and six flip modes, effectively expanding the dataset.

3.2. Visual Comparison

As illustrated in Fig. 3, we present an example to demonstrate the superiority of our method. It is evident that the lower left portion of the synthetic image is inundated with floodwaters. Nevertheless, none of the contrast methods were able to effectively extract this flood-affected area. Given the constraints of resolution, it becomes challenging to precisely discern flood-prone pixel regions from the synthetic image. However, our uncertainty-based strategy is capable of overcoming this hurdle to a considerable extent.

3.3. Quantitative Comparison

As shown in Tab. 1, we compare various methods and their respective performance metrics in terms of test accuracy and $F1$ score. Methods like UNet, HRNet, Deeplab v3+, and SegFormer achieve accuracies between 80.697% and 83.426%, indicating their effectiveness but with room for improvement. Integrating multi-scale and attention mechanisms leads to better results. Methods like UNet, UNet++, and Attention UNet with Multi-Scale Convolution Block (MSCB) show higher $F1$ scores, ranging from 84.507% to 86.033%. This suggests that leveraging multi-scale features boosts model performance. Furthermore, combining MSCB with techniques like UAFM and AUE further enhances performance, with MSCB+UAFM+AUE achieving an $F1$ score of 87.170%. The most significant improvement comes from the Multi-Fusion strategy, scoring an impressive $F1$ of 89.272%, indicating the benefits of fusing UADPNet trained on different datasets and components. Finally, applying post-processing techniques on Multi-Fusion further improves performance, achieving the highest $F1$ score of 89.843%. This underscores the importance of incorporating advanced techniques and optimizing post-processing for optimal performance in semantic segmentation tasks.

4. CONCLUSION

In this paper, we have addressed the uncertainty issues and adopted effective strategies at both the data and model levels, leading to high-precision and low-uncertainty flood mapping. As a result, we achieved first place in the 2024 IEEE GRSS Data Fusion Contest Track2. In the future, we will continue to explore how to achieve high-precision flood extraction on a global scale.

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